F70LB Life Insurance Mathematics B

Heriot-Watt University

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4 Numerical methods and continuous time

4.1 Introduction

In this chapter, we will look at numerical methods for estimating probabilities and policy values. First, we will use the Kolmogorov differential equation $\frac{d}{dt}{}_tp_x = -{}_tp_x\mu_{x+t}$ together with a numerical technique called an Euler scheme to calculate probabilities when we have estimates of the force of mortality μ_{x+t}.

Then we move on to policy values. We already have a tool to calculate policy values in discrete time. This was the recursive relationship between policy values we recapped in Chapter 2:

$$(V(t) + P_t - e_t)(1 + i_t) = q_{x+t}(S_{t+1} + E_{t+1}) + p_{x+t}\, V(t+1)$$

Therefore, in this chapter we don't look again at policy values in discrete time. Instead, we look at policy values in continuous time (where premiums and benefits are paid continuously or immediately). Our tool analogous to the recursive relationship is Thiele's differential equation. Just like with Kolmogorov's differential equation, we can use an Euler scheme with Thiele's differential equation. This will give us estimates of policy values.

4.2 Estimating probabilities in our model

Throughout this course so far we have used the future lifetime random variable T_x as our underlying model. Its distribution is completely described if we know *any one* of the following:

- the c.d.f. $F_x(t)$, or
- the p.d.f. $f_x(t)$, or
- the force of mortality μ_{x+t}.

SO if the c.d.f. $F_x(t)$ is known (meaning that it has been estimated using suitable data), then we have what we need. It can be summarised numerically by the life table l_x, which is used to compute EPVs.

However, instead we might not have estimates of the c.d.f. $F_x(t)$ of T_x available directly. We might instead have estimates of the force of mortality at all ages. We can visualise this with the following picture:

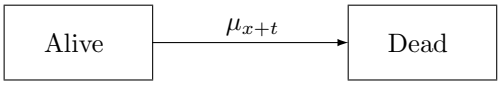


Figure 1: A model that can be used to represent a life insurance contract.

The problem then becomes: given μ_{x+t}, how can we find all the probabilities and EPVs needed to find premiums and policy values?

The answer lies in solving ordinary differential equations (ODEs). To calculate probabilities we can use the Kolmogorov differential equation $\frac{d}{dt}{}_tp_x = -{}_tp_x\,\mu_{x+t}$. combined with the initial condition ${}_0p_x = 1$.

For EPVs, we will use Thiele's differential equation discussed in section 4.6 onwards.

Very rarely, μ_{x+t} has such a simple form (e.g. a constant) that we can find analytical solutions, i.e. a simple formulae to one of both of these ODEs. Nearly always, however, we must solve them numerically using a computer. In this course, we will do this using the simplest possible method, an Euler scheme. (More sophisticated numerical methods exist.)

4.3 A comment on this approach

At first sight, this approach seems more complicated than just using a life table and, for now, it is. The payoff will come later, when we consider more complicated contracts. Here is an advance look at a model that underlies income protection insurance, in which someone who is too sick to work receives a regular income to replace their lost earnings.

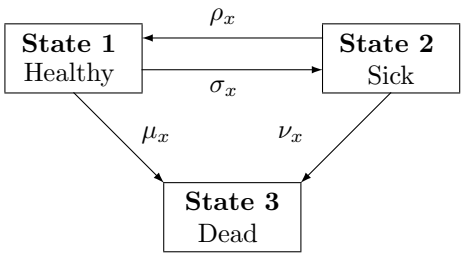


Figure 2: A model that can be used to represent a income protection insurance contract.

We *could* try to formulate this model in terms of the random times at which events take place, the analogues of the random lifetime T_x. This would be problematic. As you have seen in the Stochastic Processes course, we can set this up very nicely using transition intensities (analogous to the force of mortality). Later in this course, we will extend what you learned in Stochastic Processes by calculating EPVs (as well as probabilities).

4.4 An Euler scheme for Kolomogorov's equation

We show below how a simple recursive approach called an Euler scheme can be used to solve an ODE given a boundary condition.

An Euler scheme requires a first-order differential equation of the form

$$f'(t) = g(f(t), t)$$

and an initial condition $f(0) = c$ (or a terminal condition).

For example, put:

$$\begin{aligned} f(t) &= {}_tp_x \\ g(f(t),t) &= -f(t)\,\mu_{x+t} = -{}_tp_x\mu_{x+t} \\ f(0) &= 1. \end{aligned}$$

This sets up an Euler scheme to solve the Kolmogorov equation, conditional on the life being alive at age x.

First, we choose a suitable step size, denoted h. This should be as small as possible, consistent with the computing capacity available. In the tutorials there is an opportunity to experiment with different step sizes. For example, 0.1 year or 0.01 year might be chosen.

The Euler scheme advances the solution of the ODE in steps of length h, starting with the initial condition. It does so by assuming that the function $f(t)$ is approximately linear:

$$f(t+h) \approx f(t) + hf'(t) = f(t) + hg(f(t), t)$$

This assumption gets more reasonable as the step size decreases.

Suppose the solution has been advanced by k steps, that is, we have found approximate values of

$$f(h), f(2h), \ldots, f(kh).$$

If $f(t)$ actually were linear on $[kh, (k+1)h]$, it would be a straight line with slope $f'(kh)$, and the following would be *exactly* true:

$$f(kh + h) = f(kh) + h\, f'(kh).$$

Although $f(t)$ is not, in general, linear anywhere (certainly not in the case of the Kolmogorov equation representing human mortality) if the step size h is small enough, the error we make by assuming $f(t)$ to be *approximately* linear may be small enough to be acceptable. Therefore, we apply Equation (4.4) successively, starting with the initial condition $f(0)$. This is the Euler scheme.

4.5 Solving the Kolmogorov equation

Applying Euler's method to the Kolmogorov equation we have:

$$\begin{aligned} {}_hp_x &\approx {}_0p_x + h \left.\frac{d}{dt}{}_tp_x\right|_{t=0} \\ &= {}_0p_x + h\,[-{}_0p_x\,\mu_{x+0}] \\ &= 1 - h\,\mu_x \end{aligned}$$

and:

$$\begin{aligned} {}_{2h}p_x &\approx {}_hp_x + h \left.\frac{d}{dt}{}_tp_x\right|_{t=h} \\ &= {}_hp_x + h\,[-{}_hp_x\,\mu_{x+h}] \\ &= 1 - h\,\mu_x - h\,(1 - h\,\mu_x)\mu_{x+h} \end{aligned}$$

and:

$$\begin{aligned} {}_{3h}p_x &\approx {}_{2h}p_x + h \left.\frac{d}{dt}{}_tp_x\right|_{t=2h} \\ &= {}_{2h}p_x + h\,[-{}_{2h}p_x\,\mu_{x+2h}] \end{aligned}$$

and so on. This can easily be programmed in a spreadsheet (see tutorials), any programming language, such as Python or R, or maths software such as Wolfram Alpha. These often have functionality or packages to estimate the values of an ODE for you, but it is important to understand what they are doing.

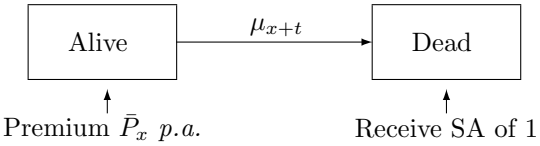
Note that the Euler scheme is the simplest numerical method for solving ODEs (which is why we use it). It is also, however, among the slowest and least accurate methods. Many better methods are available, described in any good text on numerical analysis (any maths package will certainly use one of them). We would probably not use an Euler scheme in practice.

4.6 Thiele's differential equation.

When benefits are payable immediately on death, and premiums are paid continuously, the recursive relationship between reserves becomes a differential equation, called Thiele's differential equation. We will derive it in the particular case of a whole-life non-profit insurance, issued to (x), t years ago. We assume:

- Sum assured of £1 is paid immediately on death.
- Premiums at rate $\bar{P}_x$ *p.a.* are payable continuously.

This policy can be represented as:



We suppose that the premium and valuation bases are the same, and are given by force of interest δ and force of mortality μ_{x+t}. The policy value at time t years since policy inception is again denoted $V(t)$.

[Note: In standard actuarial notation, a bar is used to denote policy values in the continuous model, e.g. ${}_t\bar{V}_x$ and ${}_t\bar{V}_{x:\overline{n}|}$. We do not use this notation.]

In the small interval of time dt:

1. The reserve earns interest of:
$$V(t)\,\delta\,dt.$$
2. Premium income is:
$$\bar{P}_x\,dt.$$

Therefore at time $t+dt$ we should have funds of:

$$V(t) + V(t)\,\delta\,dt + \bar{P}_x\,dt.$$

Applying the same reasoning as explained the recursive relationships, if all goes exactly as assumed in the bases, this should be just enough to cover the cost of paying expected death claims, and setting up the required reserve at time $t+dt$. That is, it should be equal to:

$$\mu_{x+t}\,dt + (1-\mu_{x+t}\,dt)V(t+dt).$$

Rearranging this equation, we get:

$$\begin{aligned} V(t+dt) - V(t) &= V(t)\,\delta\,dt + \bar{P}_x\,dt \\ &\quad -\mu_{x+t}(1 - V(t+dt))\,dt. \end{aligned}$$

Divide by dt to get:

$$\begin{aligned} \frac{V(t+dt) - V(t)}{dt} &= V(t)\,\delta + \bar{P}_x \\ &\quad -\mu_{x+t}(1 - V(t+dt)). \end{aligned}$$

Taking the limit as $dt \to 0$, we obtain Thiele's differential equation:

$$\frac{d}{dt}V(t) = V(t)\delta + \bar{P}_x - \mu_{x+t}(1 - V(t)).$$

The above is intuitive, not a proof. The general form of Thiele's differential equation (as seen last semester) is:

$$\frac{d}{dt}\,V(t) = \,V(t)\,\delta_t\,+\,P_t\,-\,e_t\,-\,\mu_{x+t}\,(S_t\,+\,E_t\,-\,V(t))$$

4.6.1 Proof of Thiele's equation (for a whole life policy)

We now give a proof of Thiele's equation in the special case of a whole life policy. We will need the following two results, whose proofs are tutorial questions:

$$V(t) = 1 - \frac{\bar{a}_{x+t}}{\bar{a}_x} \tag{1}$$

$$\frac{d}{dt}\left(\bar{a}_{x+t}\right) = \mu_{x+t}\,\bar{a}_{x+t} - \bar{A}_{x+t}. \tag{2}$$

Proof of Thiele:

$$\begin{aligned}
& \frac{d}{dt}V(t) \\
= \; & \frac{d}{dt}\left(1 - \frac{\bar{a}_{x+t}}{\bar{a}_x}\right) \quad \text{using (1)} \\
= \; & -\frac{1}{\bar{a}_x}\frac{d}{dt}\,\bar{a}_{x+t} \\
= \; & -\frac{1}{\bar{a}_x}\left(\mu_{x+t}\,\bar{a}_{x+t} - \bar{A}_{x+t}\right) \quad \text{using (2)} \\
= \; & -\mu_{x+t}\frac{\bar{a}_{x+t}}{\bar{a}_x} + \frac{1-\delta\,\bar{a}_{x+t}}{\bar{a}_x} \\
= \; & -\mu_{x+t}\left(1 - V(t)\right) + \frac{1}{\bar{a}_x} - \delta\left(1 - V(t)\right) \\
& \qquad\qquad\qquad\qquad\qquad\qquad \text{using (1)} \\
= \; & -\mu_{x+t}\left(1 - V(t)\right) + \frac{1-\delta\,\bar{a}_x}{\bar{a}_x} + \delta\,V(t) \\
= \; & -\mu_{x+t}\left(1 - V(t)\right) + \frac{\bar{A}_x}{\bar{a}_x} + \delta\,V(t) \\
= \; & -\mu_{x+t}\left(1 - V(t)\right) + \bar{P}_x + \delta\,V(t).
\end{aligned}$$

4.7 Estimating policy values using Thiele's equation and an Euler scheme

In our Euler scheme for a policy value, we take $f(t) = V(t)$ and Thiele's equation gives us an expression for the derivative of $f(t)$:

$$f'(t) = \frac{d}{dt}\,V(t) = V(t)\,\delta_t + P_t - e_t - \mu_{x+t}\,(S_t + E_t - V(t))$$

You can see that this is of the form we need for an Euler scheme, i.e. $f'(t) = g(f(t), t)$.

The last of the three ingredients we need to be able to apply an Euler scheme is an initial condition or a terminal condition.

Note. *The direction of time is immaterial when solving ODEs. Suppose the function $f(t)$ satisfies the ODE $f'(t) = g(f(t), t)$* and *satisfies the initial condition $f(0) = c_0$ and the terminal condition $f(T) = c_T$ for some time $T > 0$. Then we can either:*

- *choose a positive step size h and advance the solution forward from $f(0)$, or*
- *choose a negative step size $-h$ and advance the solution backwards from $f(T)$.*

In the first case, when we reach time T we should obtain the correct value $f(T) = c_T$ (to within numerical error) and, in the second case, when we reach time 0 we should obtain the correct value $f(0) = c_0$.

We can solve Thiele's differential equation either:

- forwards from the *initial condition*

$$V(0) = 0$$

  We will always have this condition when dealing with net premium policy values. More generally, this condition will hold whenever the premium has been calculated using hte principle of equivalence on the same basis (assumptions) as the policy value.

- Or backwards from *terminal conditions* like:

$$V(n) = 0 \quad \text{or} \quad V(n) = 1$$

In solving Thiele's equation, we might know an initial value, but often we do not. In fact the initial value is often the unknown quantity whose value we want to find. But we almost always know a terminal value, because:

- a policy with no maturity benefit (e.g. a term assurance) always has policy value 0 at expiry; and
- a policy with a maturity benefit (e.g. an endowment) always has policy value equal to the maturity benefit just before expiry.

Therefore, we will show the first few steps required to solve Thiele's equation backwards from such terminal conditions. In this example, we will look at a term assurance.

We use step size $-h$ for a term assurance contract sold to a life age x, with sum assured £1 payable immediately on death within n years and premium payable continuously at rate $\bar{P}$ per annum. The policy value at duration t is denoted $V(t)$ as usual.

$$\begin{aligned} f(t) &= V(t) \\ g(f(t),t) &= V(t)\delta + \bar{P} - \mu_{x+t}(1 - V(t)) \\ f(n) &= 0. \end{aligned}$$

Applying Euler's method with step size $-h$ we have:

$$\begin{aligned} V(n-h) &\approx V(n) - h \left.\frac{d}{dt}V(t)\right|_{t=n} \\ &= V(n) - h[V(n)\delta + \bar{P} - \mu_{x+n}(1 - V(n))] \end{aligned}$$

and:

$$\begin{aligned} V(n-2h) &\approx V(n-h) - h \left.\frac{d}{dt}V(t)\right|_{t=n-h} \\ &= V(n-h) - h[V(n-h)\delta + \bar{P} - \mu_{x+n-h}(1 - V(n-h))] \end{aligned}$$

and:

$$\begin{aligned} V(n-3h) &\approx V(n-2h) - h \left.\frac{d}{dt}V(t)\right|_{t=n-2h} \\ &= V(n-2h) - h[V(n-2h)\delta + \bar{P} - \mu_{x+n-2h}(1 - V(n-2h))] \end{aligned}$$

and so on.

4.8 Thiele's equation as a general tool

We have introduced Thiele's equation as a method of finding policy values. We can also use it as a tool to find premiums or even the EPV of any benefit.

4.8.1 Finding premiums using Thiele's equation

We demonstrate finding premiums using Thiele's equation by way of an example.

Example Consider a non-profit endowment with term 10 years sold to (30). The sum assured of £50,000 is payable on maturity, or immediately on earlier death. The force of mortality is μ_{x+t} and the force of interest is δ, and there are no expenses. Find the annual rate of premium $\bar{P}$.

Method 1: Set up Thiele's equation for the policy value:

$$\frac{d}{dt}V(t) = V(t)\,\delta + \bar{P} - \mu_{x+t}(50,000 - V(t)).$$

The boundary condition is $V(10) = 50,000$, but $\bar{P}$ is unknown. We can set up a calculation in Excel (or any other software) that takes $\bar{P}$ as a parameter and starts from the terminal condition $V(10) = 50,000$. Our terminal condition will always be satisfied for any value of $\bar{P}$ because of the way we've set up the calculation, but different values of the parameter $\bar{P}$ will give different values of $V(t)$ for each $t < 10$ and in particular for $t = 0$.

Note that the premium satisfies the principle if equivalence on this basis if and only if

$$\text{EPV[Benefits from time 0 onwards]} - \text{EPV[Future premiums from time 0 onwards]} = 0$$

But the left-hand side is just $V(0)$ so this is the same as saying

$$V(0) = 0$$

So to find the premium we can just use trial-and-error (the feature goalseek in Excel) to find the premium $\bar{P}$ such that $V(0) = 0$. That $\bar{P}$ is our answer.

4.8.2 Finding any EPV using Thiele's equation

Now, consider any sequence of payments and imagine a policy which makes those benefits as its payments. Take the premium to be zero. Of course, this will not satisfy the principle of equivalence and no life insurer will offer benefits for free. However, Thiele's equation still works as a calculation tool. We have:

$$\begin{aligned} V(0) &= \text{EPV[Benefits]} - \text{EPV[Future premiums]} \\ &= \text{EPV[Benefits]} - 0 \\ &= \text{EPV[Benefits]} \end{aligned}$$

Therefore, we can just set up Thiele's equation with a suitable boundary and solve as normal to find $V(0)$ which is the required EPV.

Example: Returning to our example of finding premiums for the policy in section 4.8.1, we could use Thiele's equation to find the EPV separately for a benefit of £1 (paid immediately on death) and for an annuity of £1 per annum (paid continuously while alive). Thiele's equations for these are, respectively:

$$\begin{aligned} \frac{d}{dt}V_A(t) &= V_A(t)\,\delta + 0 - \mu_{x+t}(1 - V_A(t)) \\ \frac{d}{dt}V_a(t) &= V_a(t)\,\delta - 1 - \mu_{x+t}(0 - V_a(t)) \end{aligned}$$

with boundary conditions $V_A(10) = 1$ and $V_a(10) = 0$. Then by the usual equivalence principle:

$$\bar{P} = 50,000\,V_A(0)/V_a(0).$$

In practice, we might not use Thiele's equation in this way for such a simple example since we already have ways to programme annuity and assurance calculations in a spreadsheet. However, this approach might be useful where, for example, we consider more complicated benefits or if we have Thiele's equation already set up in general and just need to vary the parameters.

4.9 Conclusion

This chapter has focussed on numerical methods for estimating solutions to the Kolmogorov and Thiele differential equations. Using an Euler scheme, as we have in this chapter, is the simplest method and more efficient and accurate approaches are available.

Among the points you take from this chapter and accompanying tutorials and labs, you should make sure that you can:

- Write down Thiele's equation for different types of policy and identify suitable boundary conditions
- Prove Thiele's equation in special cases, eg. for a whole life policy
- Write down (on paper) the first few steps of an Euler scheme for Kolmogorov and Thiele equations
- Programme Euler schemes for Kolmogorov and Thiele equations into Excel
- Use Thiele's equation to find premiums or general EPVs